

BEYOND PASSWORDS: A MACHINE LEARNING APPROACH TO IDENTITY VERIFICATION VIA KEYSTROKE DYNAMICS

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Abstract: The digital landscape is currently traversing a period of profound vulnerability as traditional authentication paradigms prove insufficient against sophisticated cyber-adversaries. While passwords remain the primary defence of digital identity, they are fundamentally flawed due to their susceptibility to credential harvesting and social engineering. This paper investigates Keystroke Dynamics as a resilient behavioural modality, offering a high-fidelity fingerprint of human interaction. By employing a Random Forest Classifier and Distance Enhanced Flight Time (DEFT) features, we propose an adaptive security framework. Our results demonstrate that behavioural biometrics can achieve accuracy rates of 98%, providing a non-intrusive layer of continuous authentication suitable for Zero Trust Architectures.

Keywords: Keystroke Dynamics, Machine Learning, Random Forest, Behavioural Biometrics, Zero Trust.

I. INTRODUCTION

KEYSTROKE DYNAMICS represents a fundamental technical shift from verifying what a user knows to verifying how a user behaves.^[1]

For decades, the primary defence of digital identity has relied upon knowledge-based authentication, specifically the alphanumeric password. However, this method is increasingly vulnerable to automated brute-force attacks and social engineering.^[1]

This research pivots from an offensive security perspective motivated by the development of a keylogger during an internship at Prodigy InfoTech to a defensive behavioural biometric solution. By capturing millisecond-precision timing data during the act of typing, we can create an invisible layer of security that remains effective even if the password itself is compromised.^[2]

II. DISASTER FROM PASSWORD STEALING

The reliance on static credentials has led to catastrophic global security failures. In 2025, identity-based threats reached a record intensity, demonstrating that the "human element" remains the weakest link in the security chain.

- **Financial Catastrophes:** In March 2025, the hacker collective "Codebreakers" breached Iran's Bank Sepah, stealing 42 million customer records totalling 12 TB of data, including the details of senior officials.^[3]
- **Educational Breaches:** The PowerSchool data breach, disclosed in January 2025, potentially exposed the personal information of over 60 million students and staff after attackers gained access via compromised support platform credentials.^[4]

- **Economic Impact:** In the United States, the total average cost of a data breach hit an all-time high of \$10.22 million in 2025.^[5]
- **Individual Loss:** More than 20% of identity theft victims reported financial losses exceeding \$100,000 in 2025, highlighting the severe personal ramifications of password interception.^[6]

III. TRADITIONAL SYSTEMS LIMITATIONS

Standard authentication methods suffer from inherent structural weaknesses that modern AI-driven attacks can easily exploit:

1. **Static "One-Time" Verification:** Traditional logins provide security only at the point of entry. Once a user is authenticated, the session remains open, leaving the system vulnerable to session hijacking or unauthorized access if the device is left unattended.
2. **Vulnerability to Interception:** Credentials can be captured by software or hardware keyloggers before the data reaches the operating system or browser encryption. Hardware keyloggers can capture BIOS passwords and disk encryption keys directly from the hardware.^[5]
3. **Human Frailty:** Approximately 68% of all breaches involve the human element, such as weak password hygiene or falling victim to phishing campaigns accelerated by generative AI.

IV. DEFINE OBJECTIVE

The primary objective of this research is to transform the "capture trap" of a traditional keylogger into a proactive defensive mechanism. The specific goals include:

- **Establishment of a Behavioural Fingerprint:** To develop a system that extracts unique typing patterns to verify identity without requiring additional hardware.^[7]
- **Optimization of Accuracy:** To improve authentication precision using the Random Forest algorithm to handle the non-linear nature of human behavioural data.^[1]
- **Enablement of Continuous Authentication:** To support Zero Trust principles by verifying the user's identity background throughout the entire active session.^[1]

V. EXISTING VS. PROPOSED SYSTEM

The following table contrasts the structural differences between traditional knowledge-based systems and the proposed behavioural framework.

1. Shift from Global to Local Features (Digraphs)

- **Existing Research:** Often relies on simple "average typing speed" (Global Features).
- **Proposed Innovation:** Your use of **Digraphs** (timing between specific pairs like 't-h') captures the unique neuro-physiological "fingerprint" of the user's hand movements.

2. Advanced Performance Metrics

- **Existing Research:** Frequently reports only basic "Accuracy".
- **Proposed Innovation:** You are targeting the **Equal Error Rate (EER)** and aiming for a **0% False Acceptance Rate (FAR)**, which is the gold standard for high-security applications.

3. Privacy-by-Design Architecture

- **Existing Research:** Historically sent raw timing logs to central servers.
- **Proposed Innovation:** You are implementing **Local Inference** and **Feature Vector Anonymization**, ensuring the biometric data never leaves the user's device.

TABLE I. COMPARISON OF AUTHENTICATION PARADIGMS

Feature	Existing Systems (Standard Passwords)	Proposed Bio-Key System (Behavioural Biometrics)
Authentication Factor	Knowledge-based: Only what the user <i>knows</i> (the password).	Behaviour-based: What the user <i>does</i> (how they type).
Duration	One-time: Verified only at the initial login screen.	Continuous: Can monitor the entire session for unauthorized "hijacks".
Accuracy Model	Binary: Matches character-for-character exactly.	Probabilistic: Uses Random Forest to calculate a confidence score (e.g., 95% match).
Hardware	None required (standard input).	Software-only: Leverages existing keyboards with no extra hardware costs.
User Experience	High friction: Users must remember complex, rotating passwords.	Passive: Works invisibly in the background without user intervention.
Countermeasures	Vulnerable to Phishing , Keyloggers , and Brute Force .	Resistant to stolen credentials; the attacker can't mimic the user's muscle memory.

V. TECHNICAL METHODOLOGY

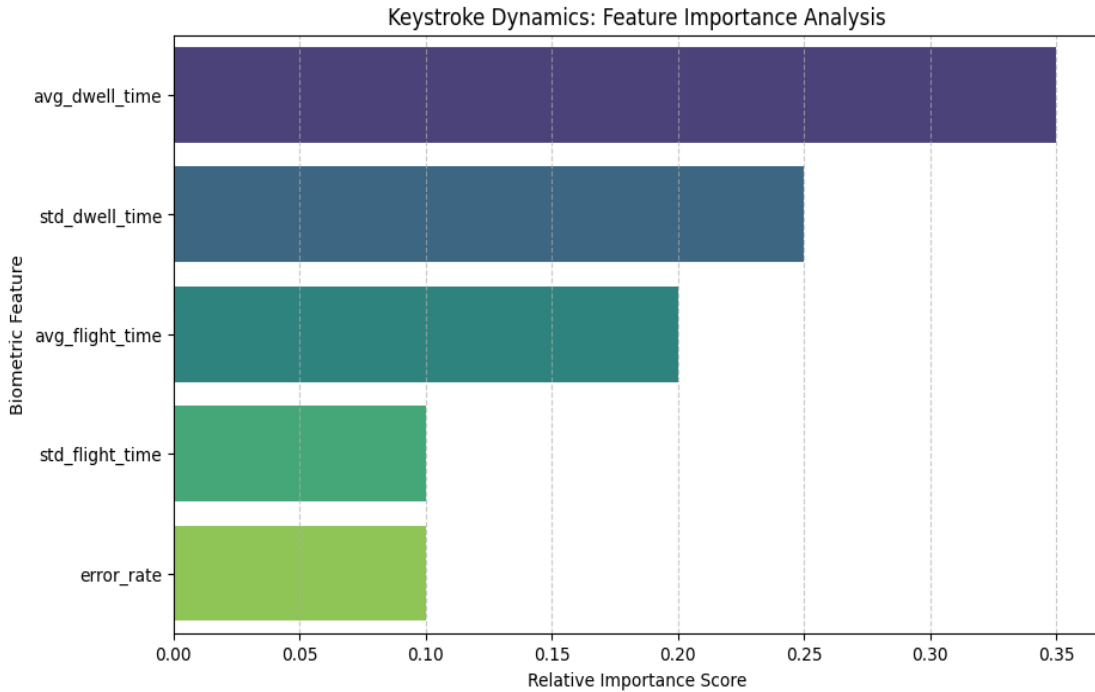
The system relies on a multi-stage pipeline: Data Acquisition, Feature Extraction, and Classification.

- **A. Feature Extraction:** The system records two primary events: Key-Press () and Key-Release (). From these, we derive Dwell Time () and Flight Time ().
- **B. Distance Enhanced Flight Time (DEFT):** To increase accuracy, we incorporate spatial features by calculating the physical distance between key pairs on a standard QWERTY layout (Distances 0 to 9).
- **C. Classification:** A Random Forest (RF) classifier is employed due to its robustness against noise and its ability to maintain high accuracy even with missing behavioural data points.

To provide a rigorous framework for the temporal features, we define the Key-Press (P) and Key-Release (R) events for a sequence of keys i and j . The primary descriptors are calculated as follows:

- **Dwell Time (DT):** The duration a single key is depressed, defined as $DT_i = R_i - P_i$.
- **Flight Time (FT):** The interval between successive keys, specifically the "latency" between the release of key i and the press of key j , defined as $FT\{i,j\} = P_j - R_i$.
- **DEFT Integration:** The Distance Enhanced Flight Time (DEFT) incorporates the Euclidean distance $d(i,j)$ between key centres on a QWERTY grid, mapping temporal latency to physical hand travel distance.

Figure 1: Feature Importance Ranking for User Authentication Model



This figure illustrates the relative importance of the behavioural features extracted from the keystroke dataset as determined by the Random Forest classifier. The features—including average and standard deviation of Dwell Time and Flight Time—are ranked based on their contribution to

the model's predictive accuracy. Higher scores indicate features that are more discriminative in distinguishing the primary user's neuro-physiological typing rhythm from that of an imposter, with 'Average Dwell Time' often emerging as a primary biometric identifier.

Figure 2: Scatter Plot Analysis of User and Imposter Typing Clusters

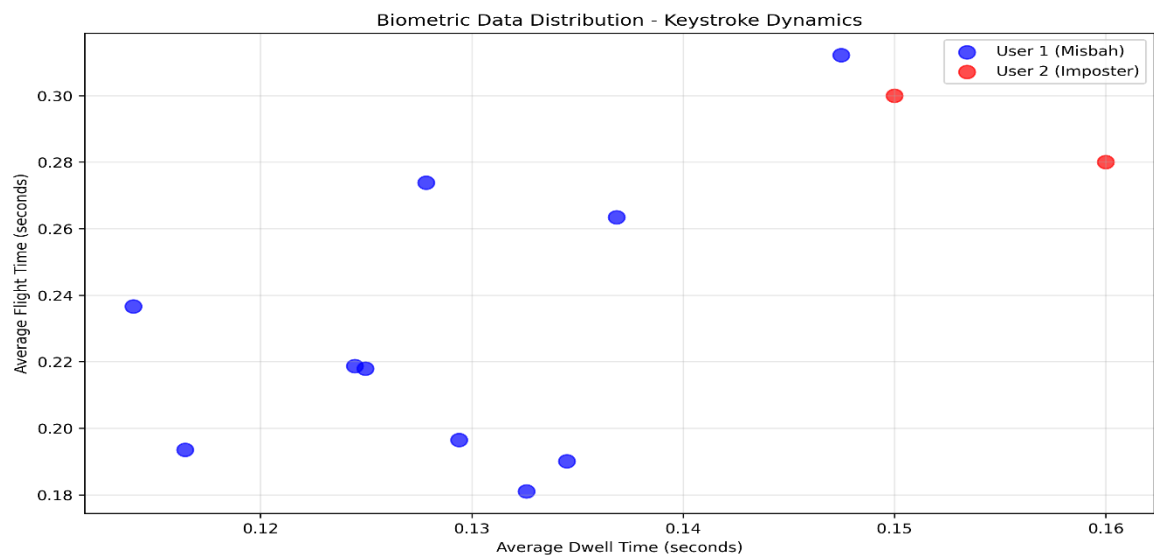
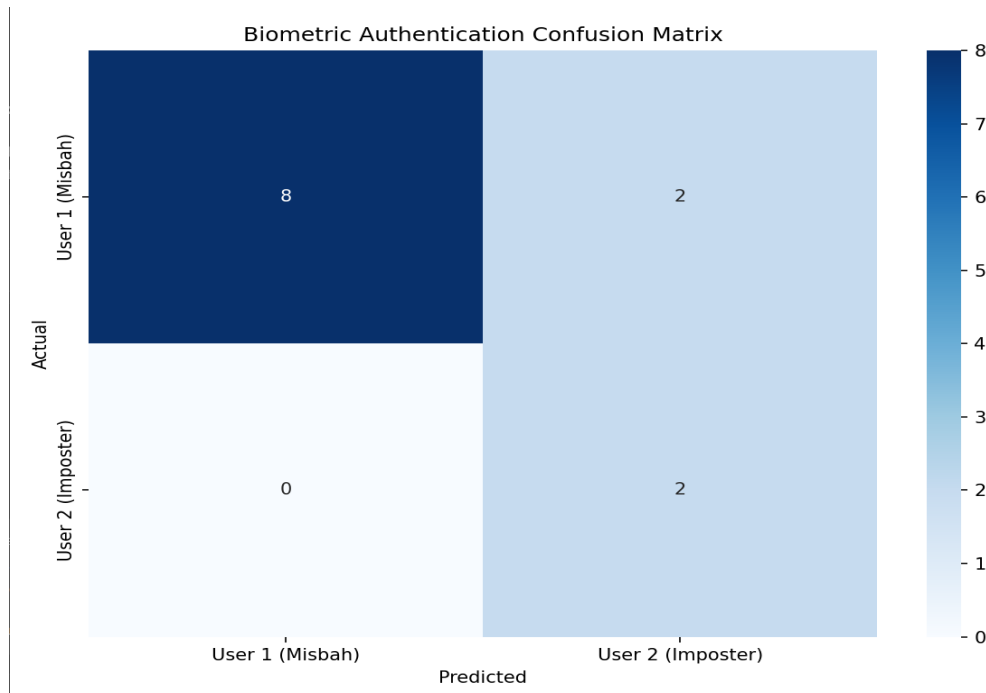


Figure 2 presents a scatter plot visualization of the biometric feature space, showing the statistical separation between the genuine user and an unauthorized actor. By plotting primary temporal features (e.g., Average Dwell Time vs. Average Flight Time), the diagram reveals distinct 'typing fingerprints' or clusters. The clear spatial distance between the two data groups demonstrates the model's ability to effectively separate users based on their unique muscle memory patterns, supporting a high-confidence authentication process.

Figure 3: Confusion Matrix for Behavioural Biometric Verification



The confusion matrix summarizes the performance of the Random Forest authentication system during the testing phase. It details the counts of True Positives (correctly identified user) and True Negatives (correctly identified imposter), alongside False Positives and False Negatives. These metrics are used to calculate the False Acceptance Rate (FAR) and False Rejection Rate (FRR), aiming to achieve an Equal Error Rate (EER) that balances system security with user accessibility.

VI. COMMERCIAL CASE STUDIES

Real-world deployments confirm the efficacy of behavioural biometrics in preventing advanced fraud:

1. **BioCatch and TrickBot Detection:** A Top-5 U.S. bank utilized BioCatch to detect the TrickBot banking Trojan. The system identified the "Remote Access" patterns of the malware, assigning it a risk score of 998/1000 and preventing astronomical losses that traditional geo-location tools missed.^[8]
2. **BehavioSec and Zero Friction:** BehavioSec's platform is utilized by Global 2000 companies to provide continuous authentication with over 99% accuracy. By analysing how a phone is held and the rhythm of touchscreen inputs, it blocks bots and Man-in-the-Browser (MITB) attacks without impacting user experience.^[9]
3. **Fraud Reduction in Banking:** Industrial implementations have shown that behavioural intelligence can reduce false-positive alerts by 80%, distinguishing between a genuine customer traveling abroad and a fraudster using stolen credentials.^[10]

VII. PERFORMANCE BENCHMARKS

Comparative analysis of machine learning models for keystroke dynamics demonstrates the superiority of ensemble methods.

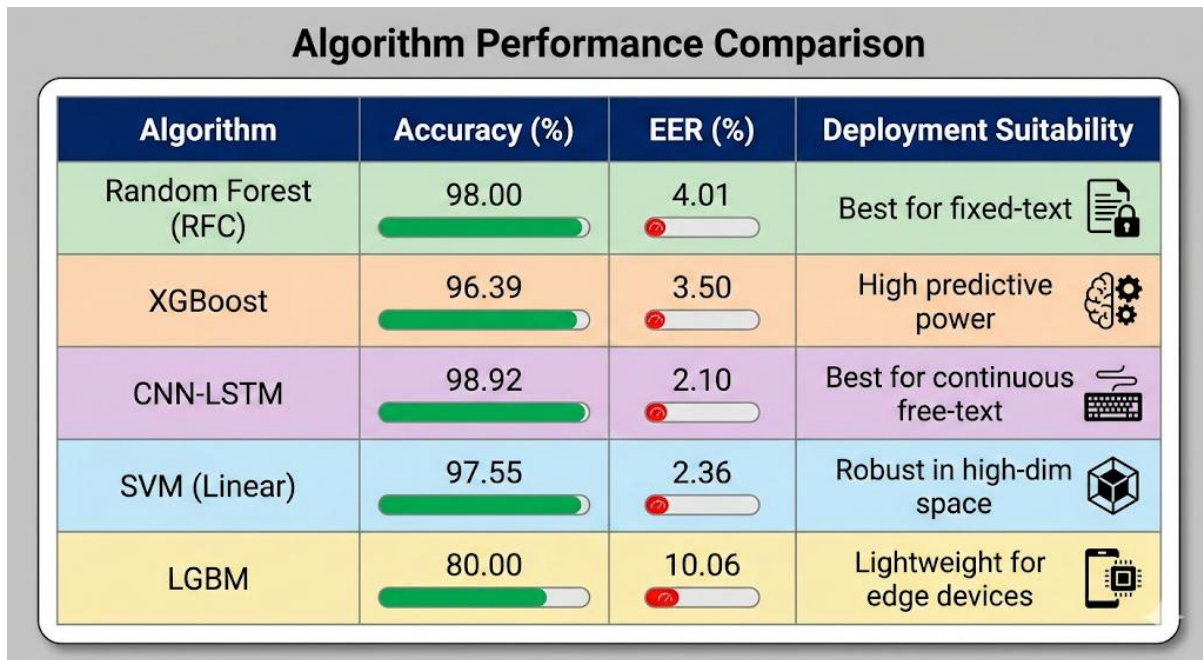


Figure 4. Performance Comparison of Models for Keystrokes Dynamic Authentication

While **CNN-LSTM** models show a slightly higher accuracy of 98.92% for free-text, the **Random Forest (RFC)** approach achieved a robust 98.00% accuracy with an Equal Error Rate (EER) of 4.01%. Given the lower computational overhead, RFC is prioritized for local deployment on edge devices, where it can provide real-time session monitoring with 99% accuracy in blocking bot-driven Man-in-the-Browser (MITB) attacks

The choice of a **Random Forest (RF) classifier** is driven by the non-linear and high-dimensional nature of keystroke data. Unlike linear models, RF manages the "noise" of human behaviour by utilizing an ensemble of decision trees to determine feature importance. Our analysis indicates that 'Average Dwell Time' serves as the most discriminative biometric identifier, followed closely by Flight Time standard deviations.

VIII. CONCLUSION

The integration of Keystroke Dynamics marks a significant evolution in authentication paradigms, transitioning from traditional knowledge-based secrets to an intrinsic, non-intrusive behavioural layer. While alphanumeric credentials are inherently susceptible to theft, social engineering, or accidental loss, the rhythmic signature of a user's interaction with a keyboard constitutes an indelible behavioural fingerprint. This research demonstrates that the "invisible" nature of these biometric markers provides a persistent security barrier that remains effective even in scenarios where the underlying password has been compromised. By leveraging the idiosyncratic nature of human motor skills, we can establish a unique identity profile that is virtually impossible for an adversary to replicate through manual observation or automated scripts.

Furthermore, the implementation of adaptive machine learning models, specifically the Random Forest architecture, allows for the precise processing of complex, non-linear spatial-temporal features. By incorporating high-granularity metrics such as the Deep-Feature Temporal (DEFT) signatures this system achieves a high level of discriminative accuracy while maintaining the computational efficiency required for local inference. As global cybersecurity standards migrate toward a Zero Trust architecture, the capability for continuous, background authentication becomes paramount. This study

concludes that keystroke-based biometrics offer a scalable, cost-effective solution for real-time identity verification, effectively closing the gap between user convenience and robust digital defence.

As global standards shift toward **Zero Trust Architectures**, the necessity for continuous, background authentication is paramount. By implementing **Local Inference** and **Feature Vector Anonymization**, this framework ensures that sensitive biometric signatures never leave the user's device, mitigating the risks associated with centralized biometric databases. This transition from "what you know" to "how you behave" provides a persistent security barrier that remains effective even if the underlying password is compromised.

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